

AIC Triad: Authentication and Authorization

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Introduction

The CIA triad is a foundational model of cybersecurity in which confidentiality, integrity, and availability constitute this model. The implementation of this model helps organizations protect information from unauthorized access, prevent improper data handling, and keep data accessible to authorized users. I think that Authentication and authorization are key in this model, as they verify identities and control access. While doing my research, I came across the 2015 Anthem data breach, which demonstrated how weaknesses in these areas can expose sensitive information on a massive scale.

The triad

In more recent times, cybersecurity has been trying to change CIA to ACI to draw a distinction between the government CIA and the cybersecurity CIA. But the framework stays the same. Now, the ACI triad provides a framework for better understanding and learning in information security. Chai (2022) highlights that the framework guides organizations and their security policies by focusing on the core principles: confidentiality, integrity, and availability. Now, confidentiality means keeping information private, with limited access to authorized users under the least-privilege principle. An example of this would be access to medical records; it should be accessed only by authorized healthcare personnel. Implementing encryption, strong passwords, MFA, and access controls are good examples of security practices that support Confidentiality.

Integrity is the means of keeping information accurate, consistent, and trustworthy. There should be some sort of chain of custody so that data should not be improperly changed, deleted, or manipulated. For example, Old Dominion University prevents unauthorized users from

changing grades in its database. There are a few things we can name that help maintain integrity. Access controls, file permissions, version control, logs, checksums, and backups can help protect data integrity and restore information if it is altered (Chai, 2022).

After we implement confidentiality and integrity, availability comes next. It ensures that authorized users can reliably access information and data systems when deemed to be.

Availability is a priority to some organizations like hospitals, financial companies, and other organizations that rely on continuous access to their data systems.

Authentication and Authorization

Doing some research, I realized that authentication and authorization are somewhat related but serve different purposes. Authentication verifies the identity of a user, device, or process before granting access (National Institute of Standards and Technology [NIST]). A common example is signing in to an account using your username, password, and MFA. So basically, authentication asks who you are, and authorization determines what an authenticated user is allowed to access or do (NIST).

Anthem Data Breach

In 2015, Anthem faced a data breach that was a clear example of a confidentiality failure. Attackers infiltrated Anthem's system via spear-phishing emails, and one employee responded to a malicious email. In turn, attackers stole electronic protected health information, or ePHI, belonging to almost 79 million individuals, including their names, Social Security numbers, medical identification numbers, physical addresses, dates of birth, email addresses, and employment information.

The impact of the breach was like no other, the information that was acquired was highly sensitive personal data that can be used for fraudulent purposes and identity theft. With that it

also resulted in major regulatory and financial consequences. Anthem had to pay \$16 million to the Office of Civil Rights and take substantial corrective action after the breach. (HHS, 2018).

Conclusion

The CIA triad provides a practical framework for protecting information and data systems. Confidentiality protects data from unauthorized access, integrity keeps information accurate, and availability keeps systems accessible to authorized users. Authentication asks the "who" question to verify identity, and authorization determines what they can access. The Anthem breach demonstrated how phishing attacks can lead to a large-scale loss of confidentiality for the organization, compromising its sensitive data. Strong authentication, access controls, regular employee training, monitoring, and risk management can help an organization prevent threats and incidents, better serving consumers and their sensitive data.

References

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